

Financial News Sentiment and Market Behavior

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Research Question and Motivation

Can news sentiment help explain S&P 500 returns and volatility?

- Financial news influences investor expectations and market reactions
- Investors continuously interpret market sentiment from headlines
- This project examines whether aggregated news sentiment is related to market behavior

Target Audience

- Traders
- Portfolio managers
- Quantitative analysts
- Market researchers

Data and Methodology

Datasets

- Financial news headlines (RavenPack)
- S&P 500 market data (FRED API)

Sentiment Scoring

- VADER sentiment analysis
- Loughran-McDonald financial dictionary

Market Variables

- Daily S&P 500 returns
- Absolute returns (volatility proxy)

Workflow

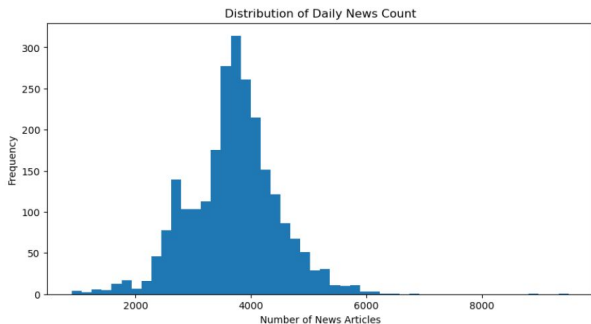


Python EDA: Understanding the Dataset

3,000–4,500

articles per trading day

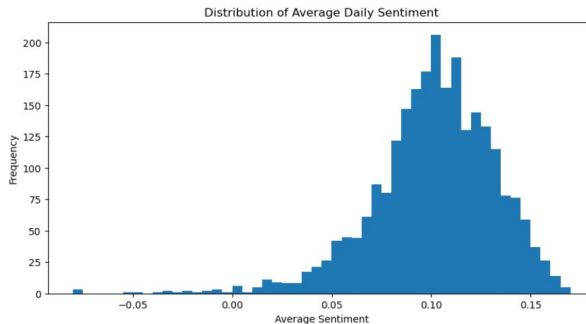
with spikes during major market events



Positive Bias

in average daily sentiment

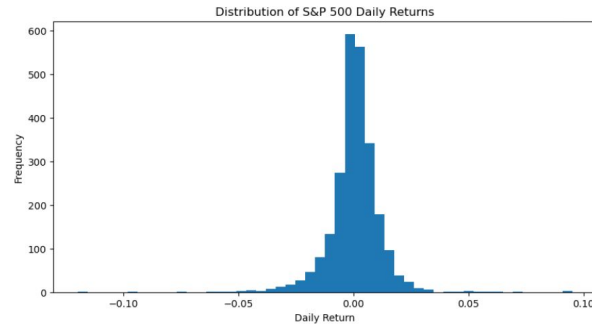
concentrated in a narrow range



Heavy Tails

in S&P 500 daily returns

reflecting occasional large market shocks



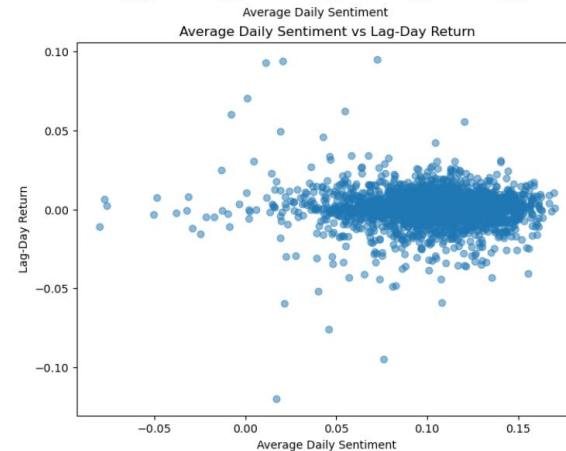
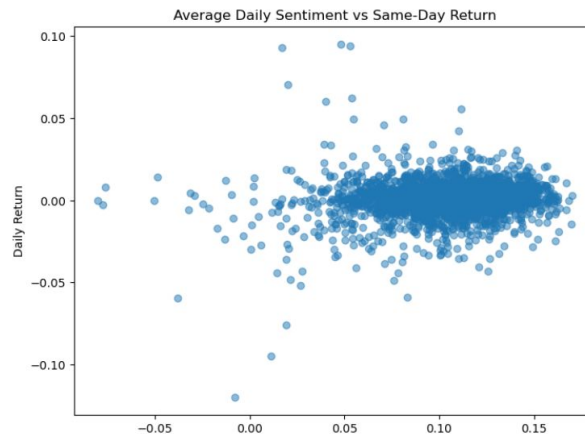
Python EDA: Sentiment and Market Returns

FINDINGS

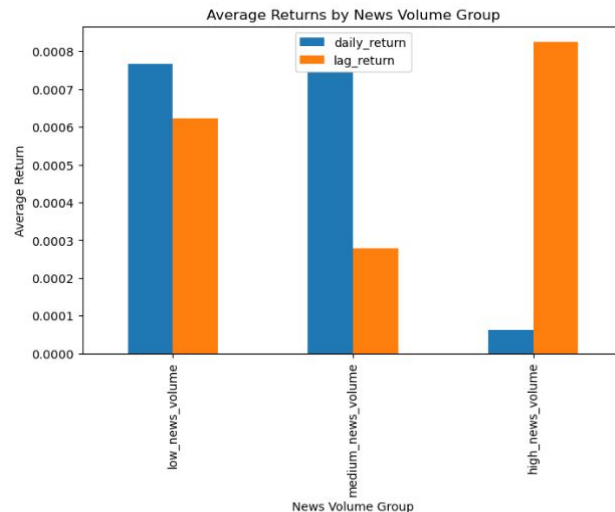
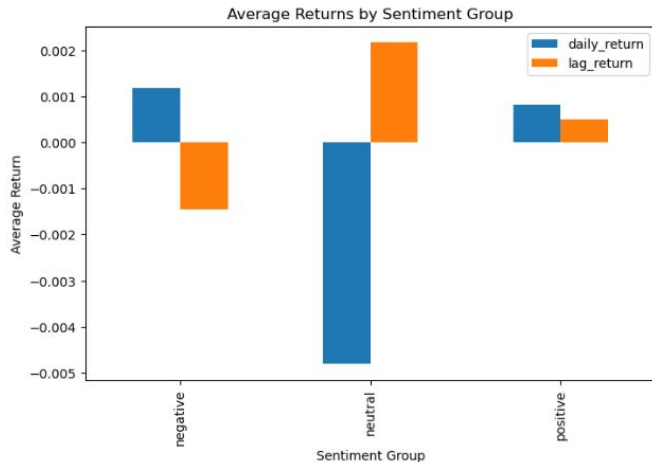
- Both same-day and lagged sentiment relationships appear weak and highly noisy.
- Similar sentiment levels often correspond to very different market returns.
- Lagged sentiment shows slightly more structured patterns than same-day sentiment.

INTERPRETATION

The relationship between same-day sentiment and market returns appears weak and highly noisy. Since market reactions may not occur immediately, delayed or cumulative sentiment effects may provide clearer market signals, which we further explore in the SQL analysis section.



Python EDA: Returns by Sentiment Group and News Volume



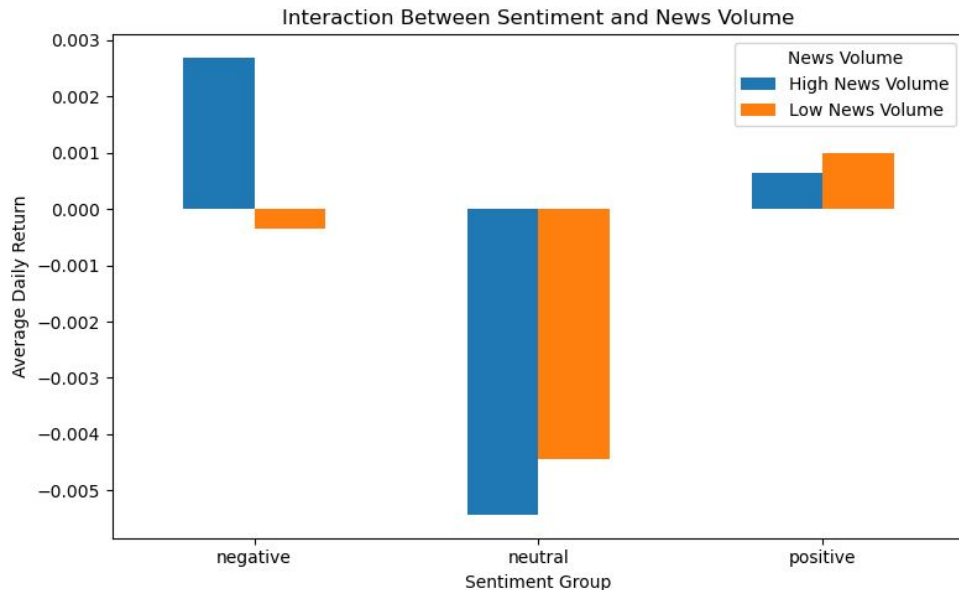
FINDINGS

- Positive sentiment is generally associated with slightly higher average returns.
- Return differences across groups are inconsistent — grouping sentiment alone reveals no stable relationship.
- High-news-volume periods show larger market fluctuations and more variable return patterns.

INTERPRETATION

The effect of sentiment may depend on overall market attention and information intensity, not sentiment direction alone.

Python EDA: Interaction Between Sentiment & News Volume



FINDINGS

- Sentiment–return relationship differs across news volume environments.
- e.g. Negative sentiment under high-news-volume behaves differently from low-volume conditions.
- News intensity influences how sentiment is reflected in market behavior.

INTERPRETATION

Further explains sentiment alone may not fully explain market reactions. The level of market attention also plays an important role.

SQL-Based Feature Engineering

Motivation: Daily sentiment appeared weak and noisy — richer temporal features were needed.

Lagged Sentiment

Previous-day trading day's sentiment

3d rolling sentiment

3-day moving average of sentiment

Absolute Return

Absolute market return — simple volatility proxy

Volatility Rank

Ranking of market volatility days

SQL Analysis: Lagged and Rolling Sentiment

Lagged Sentiment Group vs Avg Return

	lag_sentiment_group	observations	avg_lag_sentiment	avg_daily_return	avg_lag_return	avg_abs_return
0	Low Lagged Sentiment	824	0.068368	0.000484	0.001014	0.008368
1	Medium Lagged Sentiment	824	0.103569	0.001046	0.000382	0.006763
2	High Lagged Sentiment	824	0.132249	0.000045	0.000330	0.006866

Rolling Sentiment Group vs Avg Return

	rolling_sentiment_group	observations	avg_rolling_sentiment	avg_daily_return	avg_lag_return	avg_abs_return
0	Low Rolling Sentiment	825	0.075205	0.000306	0.000857	0.008415
1	Medium Rolling Sentiment	824	0.103067	0.000496	0.000437	0.006654
2	High Rolling Sentiment	824	0.125994	0.000774	0.000432	0.006927

FINDINGS

- Higher lagged sentiment does not consistently lead to higher future returns.
- Rolling sentiment shows a clearer pattern — returns increase from low to high sentiment periods.
- Low rolling sentiment periods exhibit higher volatility (higher absolute returns).

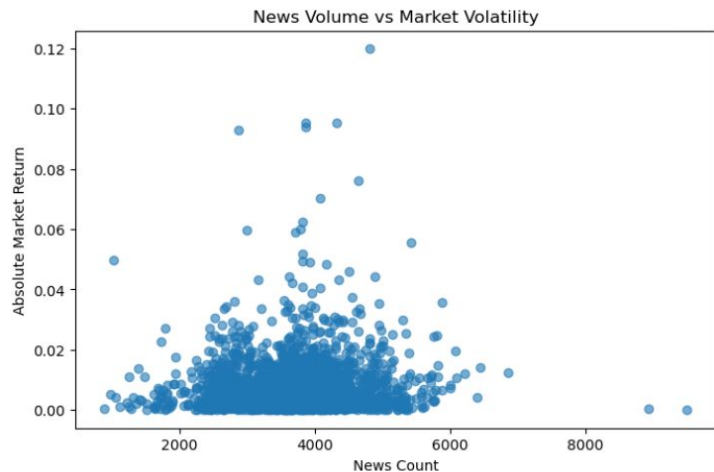
INTERPRETATION

Temporal features provide more structure than raw daily sentiment, but the relationship remains market-condition dependent.

SQL Analysis: News Volume and Market Volatility

Avg Returns and Volatility by News Volume Group

	news_volume_group	observations	avg_news_count	avg_daily_return	avg_lag_return	avg_abs_return
0	low_news_volume	827	2876.627570	0.000768	0.000622	0.006728
1	medium_news_volume	822	3728.965937	0.000746	0.000278	0.007456
2	high_news_volume	824	4498.635922	0.000062	0.000826	0.007817



FINDINGS

- High-news-volume periods exhibit the highest average absolute returns — indicating greater volatility. (Same finding as python EDA)
- Average daily returns remain similar across volume groups.
- Extreme volatility events tend to occur during elevated news activity.

INTERPRETATION

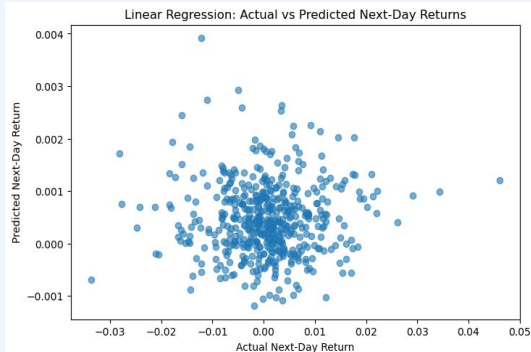
News volume appears more strongly linked to market uncertainty than to directional return prediction. Heavier news flow coincides with greater investor uncertainty.

Predictive Models

Linear Regression

- R^2 : -0.0033
- MAE: 0.006413
- RMSE: 0.008822

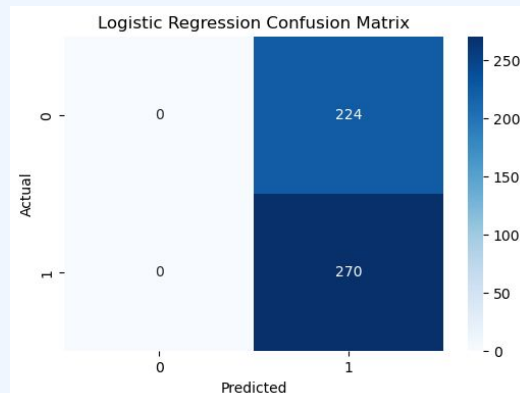
Near-zero R^2 indicates sentiment features explain very little return variance.



Logistic Regression

- Accuracy: 54.7%

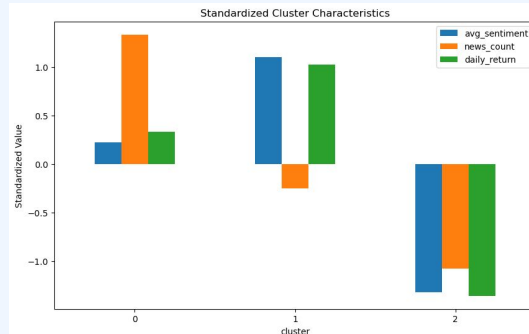
Barely above random (50%) — sentiment has limited directional signal.



K-Means Clustering

- Features: avg_sentiment, news_count, daily_return, positive_share, negative_share, neutral_share

Used to identify market regimes from combined sentiment and return features.



All three models confirm that news sentiment alone offers limited predictive power for market returns.

Conclusion

Key Findings

- Financial news sentiment alone provides limited predictive power for S&P 500 returns.
- Same-day sentiment relationships are weak and highly noisy; temporal features (lagged and rolling sentiment) provide slightly clearer patterns, but results remain inconsistent.
- News volume shows a stronger relationship with market volatility than with return prediction.

Overall Interpretation

Market reactions appear to depend not only on sentiment direction, but also on market attention and information intensity. Financial news may be more useful for understanding market uncertainty and volatility environments than for predicting exact market direction.

Future Improvements

- Apply finance-specific NLP models
- Incorporate macroeconomic and event-based variables